

CBSE Practice Worksheet

Class 6 | Science | Difficulty: Hard | Set B

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. Which law relates induced EMF to rate of change of magnetic flux?
(A) Lenz's law
(B) Faraday's law
(C) Ampere's law
(D) Ohm's law
2. What is the escape velocity from Earth's surface?
(A) 8 km/s
(B) 11.2 km/s
(C) 15 km/s
(D) 25 km/s
3. Which reaction is endothermic?
(A) Combustion
(B) Neutralization
(C) Photosynthesis
(D) Respiration
4. What is the hybridization of carbon in methane?
(A) sp
(B) sp^2
(C) sp^3
(D) sp^3d
5. What is the coordination number in FCC structure?
(A) 6
(B) 8
(C) 12
(D) 14
6. What is the magnetic quantum number for 3d orbital?
(A) -2 to +2
(B) -3 to +3
(C) -1 to +1
(D) 0 only
7. What is the wavelength of Lyman series limit?
(A) 91.2 nm
(B) 121.6 nm
(C) 656.3 nm
(D) 364.6 nm
8. What is the efficiency of an ideal Carnot engine operating between 300K and 400K?
(A) 25%
(B) 33%
(C) 50%

(D) 75%

9. Which type of mutation involves change in single nucleotide?

- (A) Deletion
- (B) Insertion
- (C) Point mutation
- (D) Frameshift

10. What is the wavelength of de Broglie wave for an electron?

- (A) $\lambda = h/mv$
- (B) $\lambda = mv/h$
- (C) $\lambda = hv/m$
- (D) $\lambda = m/hv$

Section B: Fill in the Blanks (1 mark each)

- 11. The process of splitting water during photosynthesis is called ____.
- 12. The Krebs cycle occurs in the ____ of mitochondria.
- 13. The band gap in semiconductors is typically ____ eV.
- 14. The uncertainty principle was given by ____.
- 15. For a reversible process, entropy change of the universe is ____.
- 16. The IUPAC name of $\text{CH}_3\text{CH}_2\text{CHO}$ is ____.
- 17. The magnetic field inside a solenoid is given by $B = \mu \blacksquare$ ____.
- 18. In SN_2 reactions, the rate depends on both ____ and nucleophile.

Section C: Short Answer Questions (2 marks each)

- 19. Derive the expression for electric field due to a dipole.
- 20. Describe electromagnetic induction and derive Faraday's law.
- 21. Describe the light-dependent and light-independent reactions of photosynthesis.
- 22. Explain Le Chatelier's principle with examples.
- 23. Derive the lens formula and explain its applications.
- 24. Explain the structure and working of a nuclear reactor.
- 25. Describe the mechanism of nerve impulse transmission.

--- End of Worksheet ---

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